

Nour Mahmoud

Raleigh, NC

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Summary

CS student at NC State (4.0 GPA, May 2026) with hands-on experience in full-stack development, HCI research, and teaching. Skilled in Java, Python, C/C++, C#, SQL, and JavaScript. Published researcher (HCII). Seeking a Software Engineer New Grad role.

Education

North Carolina State University | Raleigh

May 2026

Bachelor of Computer Science | GPA: 4.0 / 4.0

Experience

Research Assistant, NSF REU Program (HCI) | Internship | Raleigh

May 2025 - Present

- Conducted research into collaborative coding, annotating 2,700 dialogue turns from ten student teams using Python & Jupyter; achieved inter-rater agreement (Jaccard similarity > .80)
- Uncovered patterns in decision-making tied to reduced debugging time, through statistical interpretation and iterative labeling
- Co-authored a research paper on findings; *accepted into the HCII 2026 Conference*

Computer Science Ambassador | Volunteer | Raleigh

Aug. 2024 - Present

- Represented the CSC department in outreach, campus tours, and events; communicated complex technical topics to prospective students & families
- Coordinated with faculty/staff and peer teams to plan and execute events smoothly

North Carolina State University | Teaching Assistant (TA) & Senate Member | Raleigh

Aug. 2023 - Present

- Mentored 400+ students in Intro to Java and Data Structures & Algorithms; elected to TA Senate, conducted lab observations to evaluate TA practices, and participated in behavioral interviews to select and place new TAs
- Guided students through in-depth debugging and complex concepts such as recursion, sorting algorithms, and data structure implementation
- Provided clear, rubric-based feedback to strengthen problem-solving and coding practices
- Collaborated with faculty to refine instruction and promote inclusive, supportive learning environments

Projects -- ([View full portfolio for more](#))

Wolfcafe Full Stack Web Application | Java, Javascript, Spring Boot, MySQL, REST, Maven, Hibernate

- Led frontend development and coordinated backend design on a 6-person agile team to deliver a production-style cafe management and e-commerce platform
- Implemented a RESTful API with Spring Boot and Hibernate ORM; designed 5-role access control (Admin, Manager, Barista, Customer, Guest) and JWT-based authentication
- Built React UI with role-specific navigation, recipe variants (S/M/L), order pickup by number, guest checkout, and order history; integrated Vitest and React Testing Library for automated frontend tests
- Drove agile practices: GitHub issues/PRs, story points, sprint reviews, system test plans, and unit tests; applied layered architecture (Controller → Service → Repository) for maintainability

SDL2 Game Engine: Multiplayer Physics & Collision | C++, SDL2, ZeroMQ, Multithreading

- Led a 3-person team to design and build a 2D game engine from scratch with a component-entity model, collision pipeline, and multithreaded update loop over 4 months
- Implemented a ZeroMQ (Router/Dealer) networking layer for ordered client-input broadcast and explicit disconnect handling to support multiplayer gameplay
- Delivered two complete games (Bubble Shooter, Space Invaders) on the custom engine with stable 60 FPS on mid-range hardware
- Architected collision and decoupled gameplay code for fast iteration and reuse

Technical Skills

- **Programming Languages:** Python, Java, SQL, JavaScript, C++, C, C#
- **Frameworks & Tools:** React, Spring, Hibernate, Tomcat, Vite, ZeroMQ, JUnit, Maven
- **Concepts:** Multithreading, Networking, A* Search, REST APIs, Iterative Project Management
- **Environments:** Unix/Linux, Git & GitHub, Jupyter, VS Code, Eclipse, CUDA, Unity, Unreal Engine, Godot